IMPERIAL

TropiCycloneNet Basin Generalisation

Physics-guided Domain Generalisation For Tropical Cyclone Forecasting

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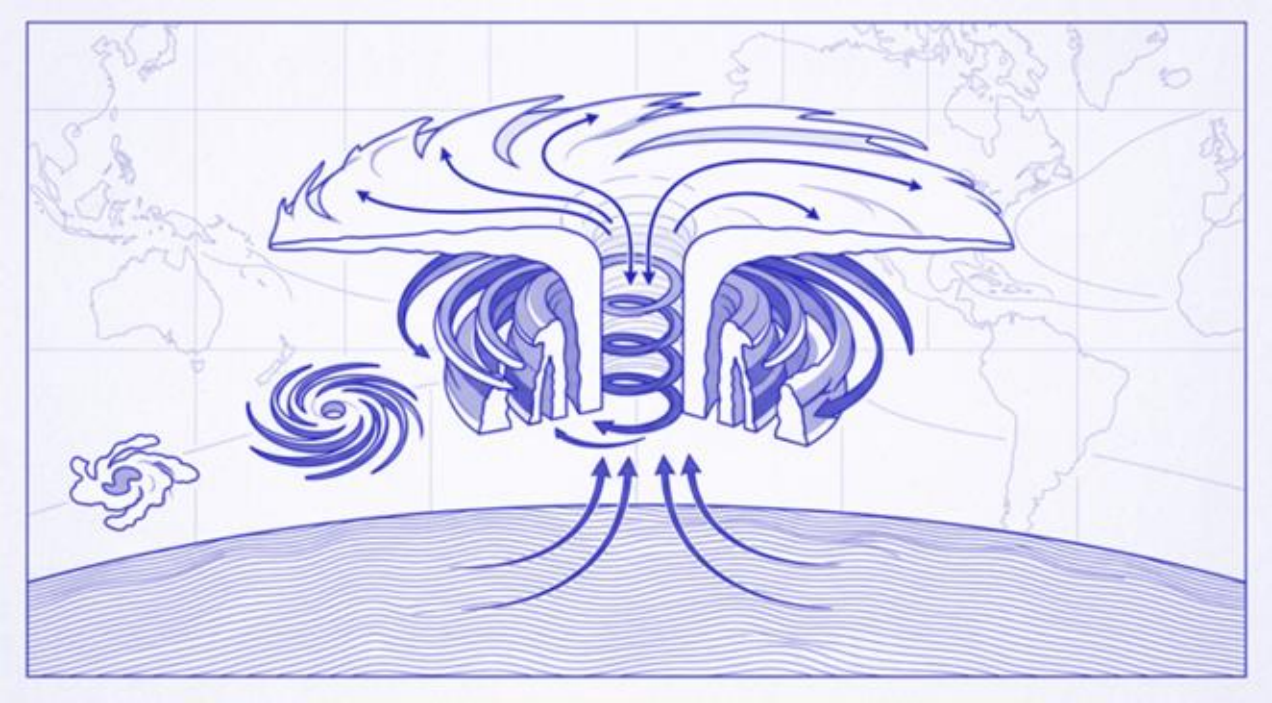
The Threat Of Tropical Cyclones

Introduction

Tropical Cyclones are intense, rotating weather systems fuelled by warm ocean waters, characterized by low atmospheric pressure, strong winds, and heavy rain.

Why This Work Matter?

- TCs are among the most destructive natural hazards on Earth, causing massive loss of life and severe economic damage to coastal communities worldwide.
- Accurate forecasting is critical for early warnings and disaster preparedness. However, creating a unified global prediction system remains a major scientific hurdle.



Tropical cyclones cause $51 billion in annual economic losses globally and affect over 20 million people each year.

Rapid Intensification, defined as a wind speed rise of 30 kt in 24 hours, remains the hardest phase to forecast and is increasing in frequency under climate change.

Why Basin Generalization?

Motivation

Distinct Climatology

The Issue
Models trained in one ocean basin frequently fail in others due to differing synoptic drivers (monsoon troughs, Coriolis gradients, subtropical highs).

The Impact
Tropical cyclones cause $51 billion in annual economic losses globally and affect over 20 million people each year.

WMO, 2022

Performance Gap

The Issue
The performance gap is driven by covariate shift, SST climatology, vertical wind shear profiles, and ocean heat content distributions differ substantially across basins

The Impact
The model cannot generalize to a completely unseen target basin without retraining.

Giffard-Roisin et al., 2020

High-Stakes Failure Modes

The Issue
Rapid Intensification (RI) events are rare but dangerous.

The Impact
Poor cross-basin transfer leads to dangerous mispredictions of RI.

DeMaria et al., 2021

Where Our Work Fits

Filling The Gap

Theory VS Practice

Moving from basin-specific models to unified forecasting that can generalize.

The Dataset

We use TropiCycloneNet Dataset, providing standardized, multi-modal records.

The Methodology

We propose a novel physics-guided constraint for domain generalization.

TropiCycloneNet (TCND)

Dataset

Massive Temporal Scope

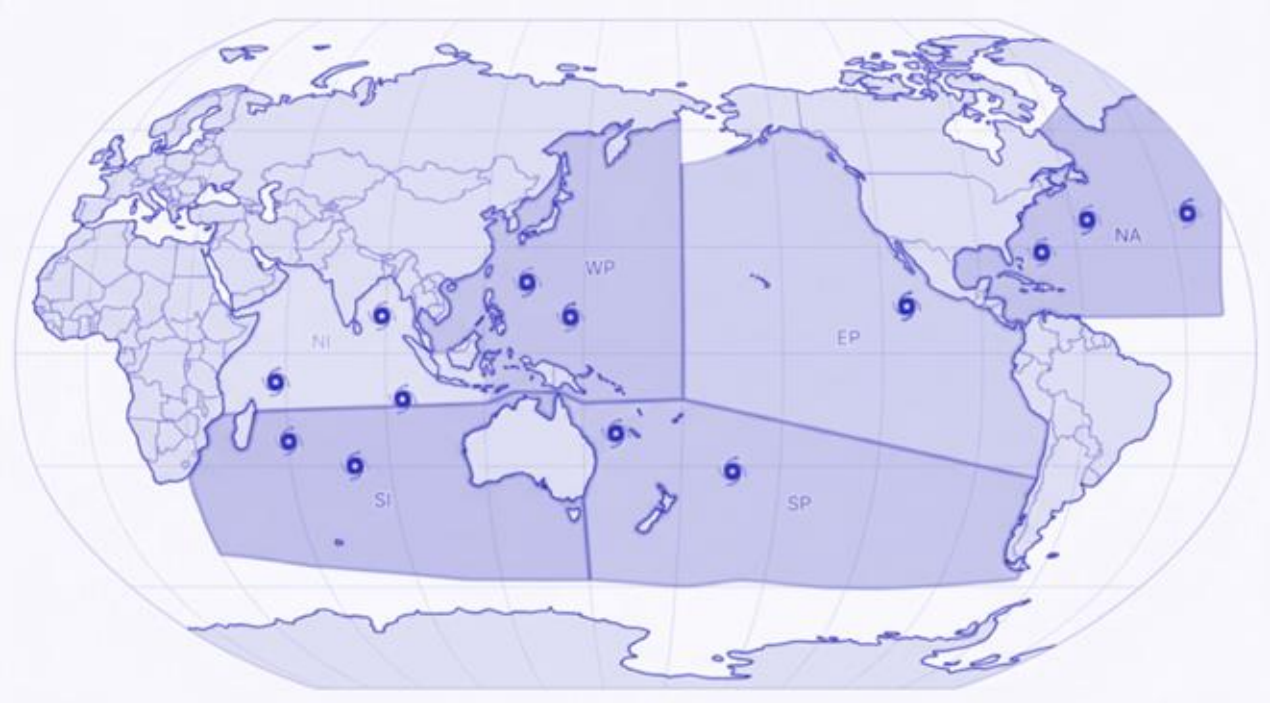
Contains records spanning over 70 years (1950–2023), encompassing more than 3,720 distinct tropical cyclones.

Six Distinct Ocean Basins

Covers major genesis regions - Western Pacific (WP), North Atlantic (NA), Eastern Pacific (EP), North Indian (NI), South Indian (SI), and South Pacific (SP).

Multi-Task Prediction

Helps us in evaluating four tasks ahead in time: track direction, direction change classification, intensity change classification, and continuous intensity regression.



TropiCycloneNet (TCND)

Dataset

Data1D
Inherent Attributes

Captures the track and intensity history of the storm.

(8 columns per 6-hour step)

Col	Field Name	Description
0	ID	Time step counter
2	LAT_norm	Normalized latitude
3	LONG_norm	Normalized longitude
4	**WND_norm**	**Max wind (Reg Target)**
5	**PRES_norm**	**Min pres (Reg Target)**
6	YYYYMMDDHH	UTC timestamp
7	Name	Storm identifier

Data3D
Meteorological Grid

ERA5 reanalysis patches (20°×20°), 81×81 resolution.

(U/V-wind, GeoP. Height & SST)

Channel	Variable	Levels (hPa)
0–3	U-wind	200, 500, 850, 925
4–7	V-wind	200, 500, 850, 925
8–11	Geopotential Z	200, 500, 850, 925
12	SST	Surface

Env-Data
Environmental Context

Month, Area, Velocity & Location bins.

(94-dim pre-computed vector)

Slice	Field	Encoding
[0:12]	Month	one-hot (12)
[12:18]	Area	one-hot (6)
[26:74]	Lat/Lon Bins	one-hot (48)
[74:94]	History Track	one-hot (20)
Labels	**Class Targets**	Int & Dir (+24h)

Generalization Pipeline

Training Flow

1
Feature Selection
Filters Inputs Based on cross-basin stability.

→

2
Basin Selection
Greedy Search for optimal source basins.

→

3
Model Training
Training multiple models, along with PhysIRM.

→

4
Target Evaluation
Comparing performance against target basin.

A sequential decision pipeline moving from data analysis to robust evaluation

Source Basins
WP, NA, EP, NI, SP
Selected Basins For Training

→

Model
Multimodal Backbone
Domain Generalization On Source Basins

→

Held-Out Target
SI
Zero-shot Evaluation On Unseen Basin

Leave-One-Basin-Out Strategy

Domain Generalization Baselines

Modelling

ERM
Empirical Risk Minimization

The standard baseline

IRM
Invariant Risk Minimization

Gradient penalty

V-REx
Variance Risk Extrapolation

Variance penalty

CORAL
Correlation Alignment

Aligns features stats (e.g. cov. matrices)

DANN
Domain-Adversarial Neural Network

Learn domain invariant features

MAML
Model-Agnostic Meta-Learning

Learn transferable and robust features

All methods share the exact same multimodal backbone and task heads for a fair comparison.

PhysIRM

Physics-Guided Generalization

Instead of forcing the entire representation to be invariant, can we isolate invariance to the physical mechanisms that govern the cyclones everywhere?

Generic Invariance

Over-regularization
Forces the entire feature space to be invariant across basins.

Collapses Context
Erases local environment signals necessary for tasks like track direction prediction.

Black-box
Learns arbitrary invariances that lack physical interpretability.

VS

PhysIRM (Proposed)

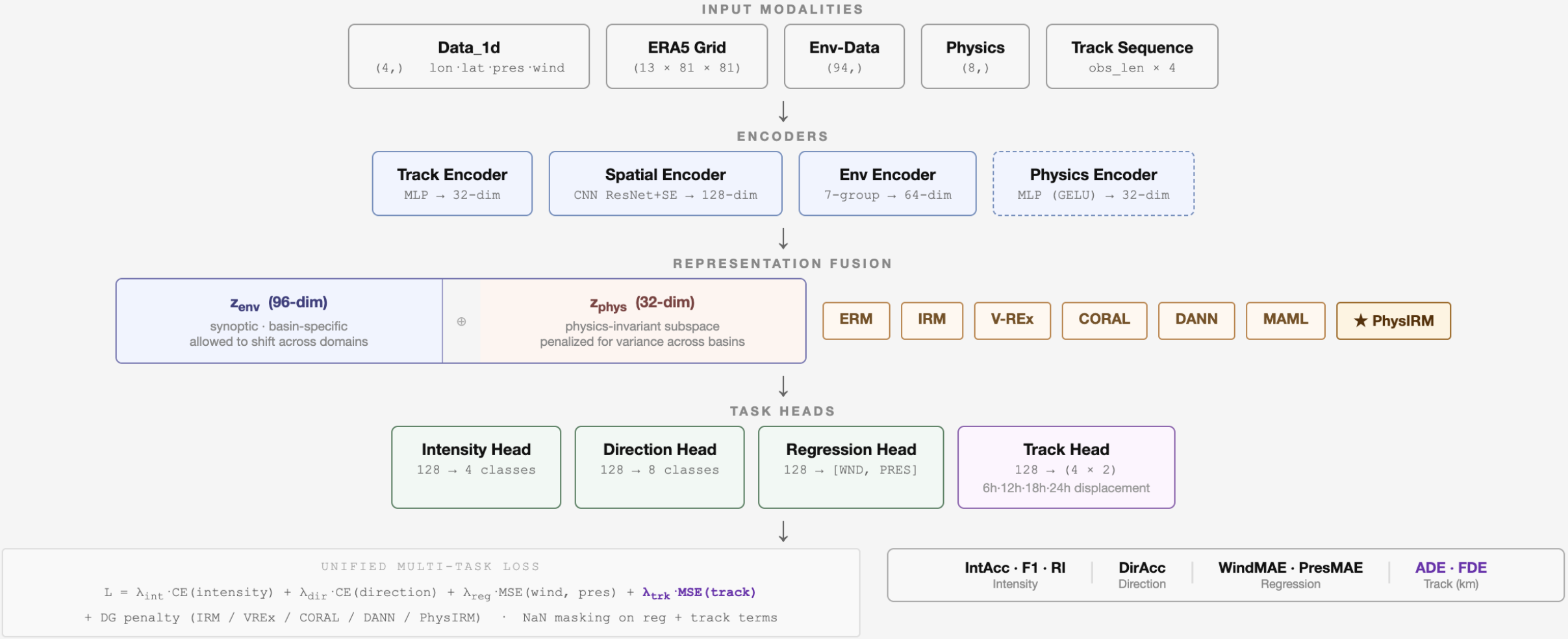
Targeted Constraints
Restricts invariance penalty to the physics sub-space only.

Preserves Context
Allows the environment sub-space to shift, maintaining accuracy on directional tasks.

Grounded
Explicitly anchors the invariant space to analytical thermodynamics (SST, shear).

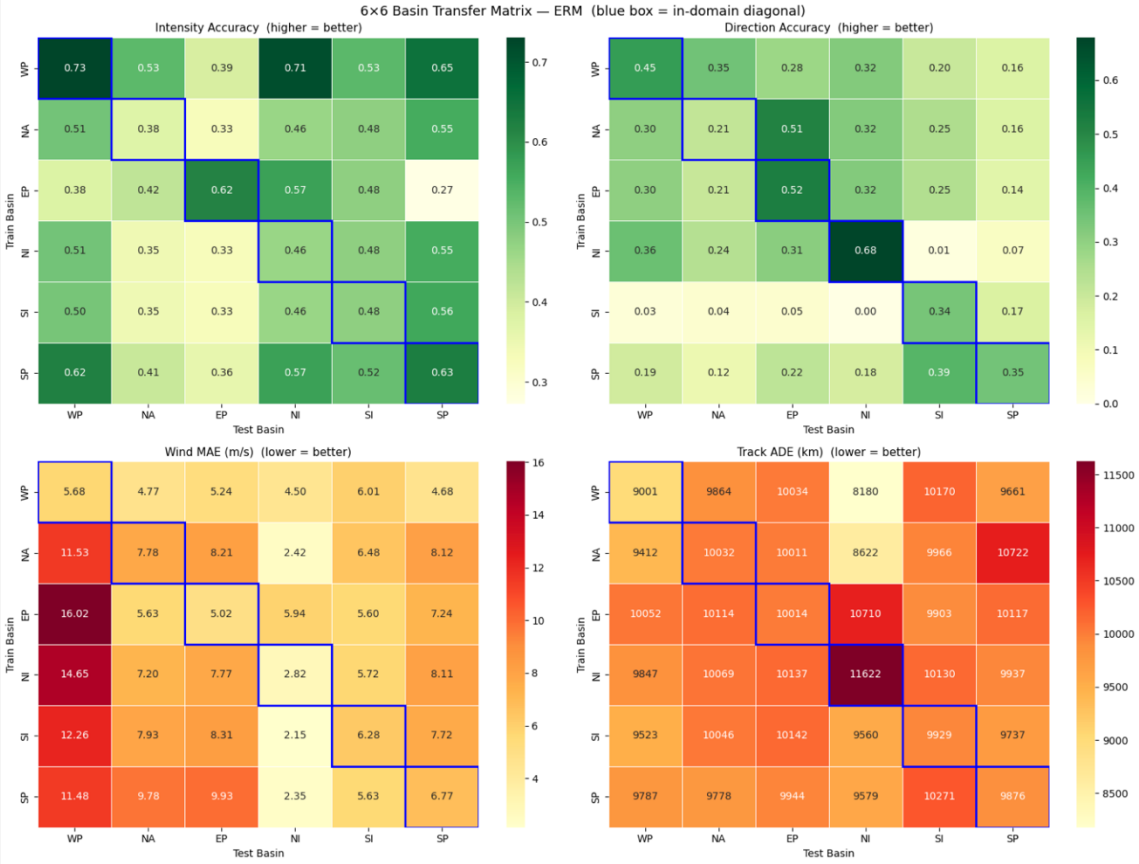
Decompose the feature space into z_{phys} (thermodynamic invariants) and z_{env} (synoptic shifts).
Apply IRM penalty only on z_{phys}.

Our Architecture
Version 1



Basin Transfer Matrix

Findings

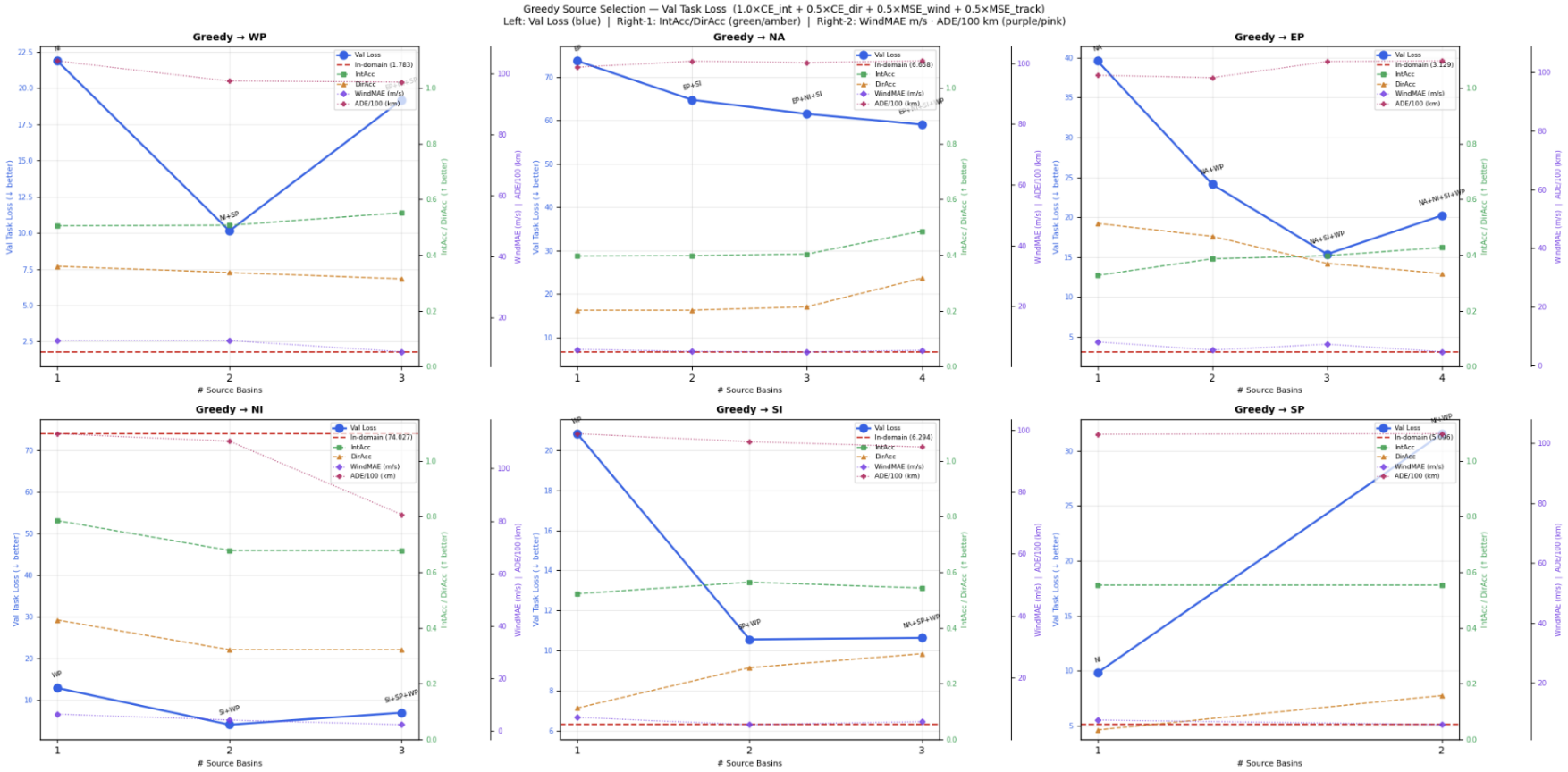


- In-domain training is not always optimal, NI, NA, and EP all have at least one cross-basin source that beats their own in-domain score.
- NI is the biggest beneficiary of transfer, In-domain NI = 0.464 (only 28 test samples). WP as source jumps it to 0.714, a 54% gain, the largest improvement in the entire matrix.
- Track ADE is not yet meaningful, All ADE values are in the 8,000–11,000+ km range, indicating the track head hadn't converged at this stage.

Greedy Source Selection

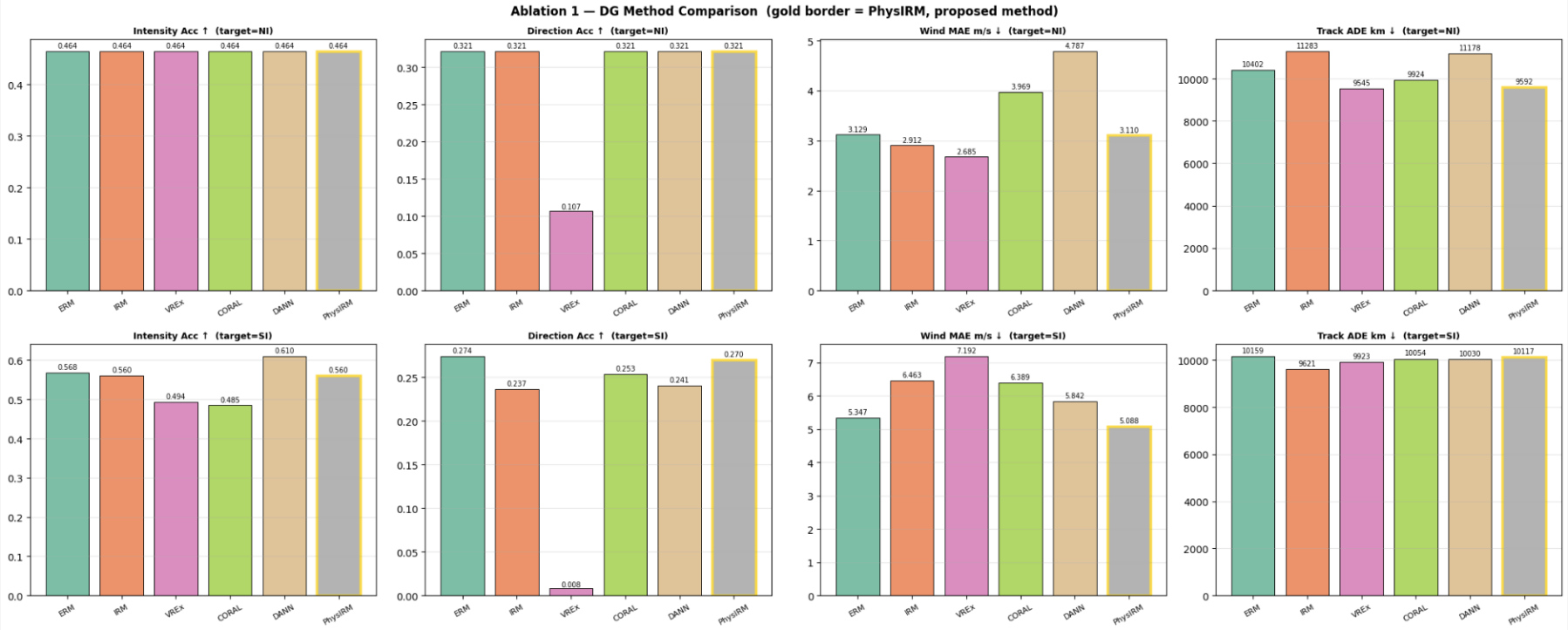
Findings

- WP is the universal anchor, WP appears in the optimal source set for 5 out of 6 targets (NA, EP, NI, SI, SP)
- 2–3 sources are enough, more hurts. The greedy algorithm stops at 2–3 basins for every target.
- Greedy beats in-domain for data-scarce targets, NI greedy (SI+SP+WP) achieves 0.679 IntAcc vs 0.464 in-domain (+46%).
- Geography is not destiny, NI's best sources are SI+SP+WP, two southern hemisphere basins. EP's best set includes NI and SI alongside NA. The greedy search overrules geographic intuition; atmospheric dynamics matter more than basin proximity.



Ablation Study - 1

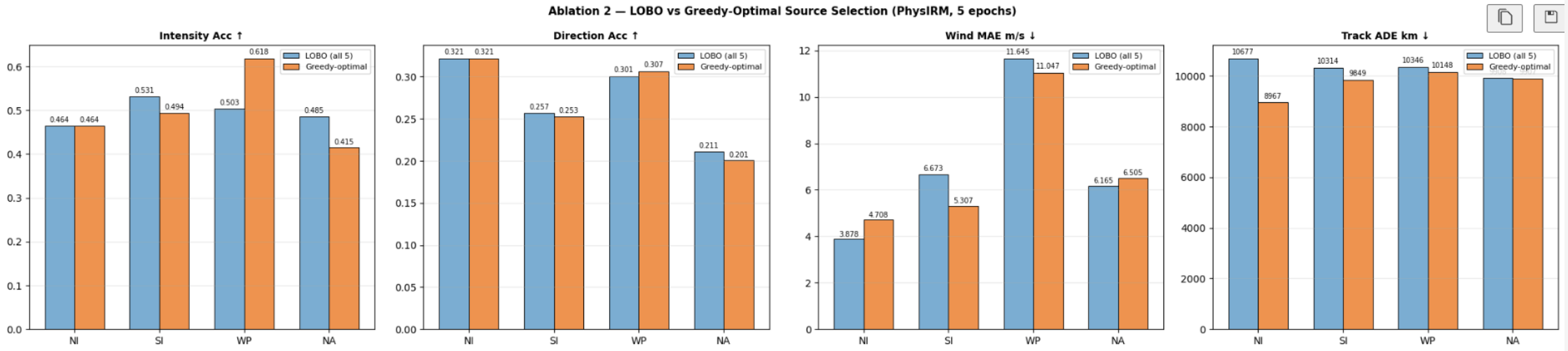
Findings



- **NI is floor-locked:** all methods tie at 0.464 IntAcc, Only 28 test samples means every method hits the majority-class ceiling. No DG algorithm can overcome data scarcity.
- **DANN wins on classification (SI IntAcc= 0.610):** Domain-adversarial training forces the model to learn basin-invariant features, giving the best intensity accuracy. But it sacrifices regression quality (WindMAE = 5.84 m/s, worst among competitive methods).
- **PhysIRM wins on regression (SI WindMAE = 5.09 m/s):** Physics constraints anchor the invariant subspace to physically meaningful features, giving the best wind error without collapsing direction. Most balanced method overall.

Ablation Study - 2

Findings



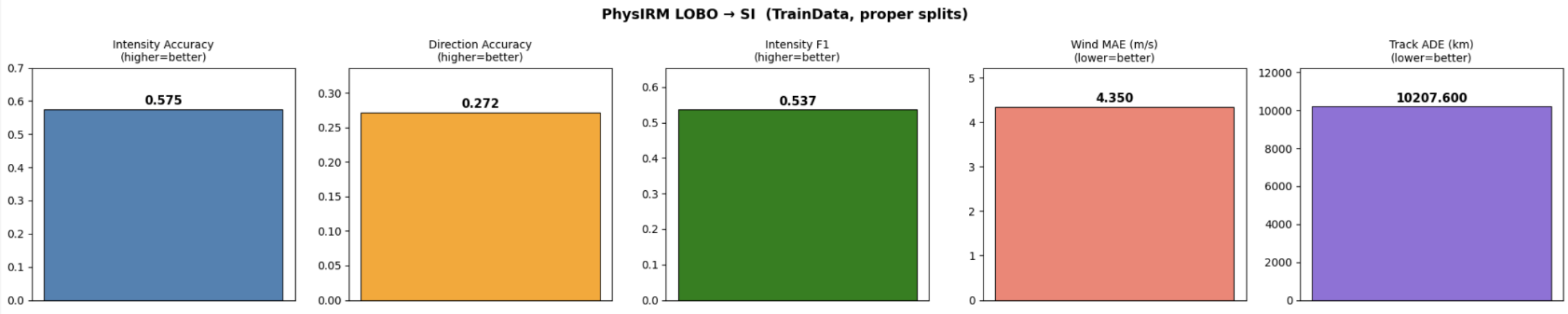
- **LOBO beats Greedy for 3 out of 4 targets:** For SI (−3.7%) and NA (−7.0%), using all available source basins wins over the hand-picked greedy subset. More source diversity is better when the target.

- **Greedy wins decisively for WP (+11.5%):** Greedy subset EP+NI+SP achieves 0.618 vs LOBO's 0.503. Removing NA and SI avoids redundant or noisy signal, WP benefits from a compact, non-overlapping source set rather than throwing all data in.

- **NI is again floor-locked:** source composition is irrelevant, LOBO (0.464) and Greedy (0.464) are identical on IntAcc. Only ADE differs marginally. Confirmed: NI's 28 test samples make source selection a non-factor for classification.

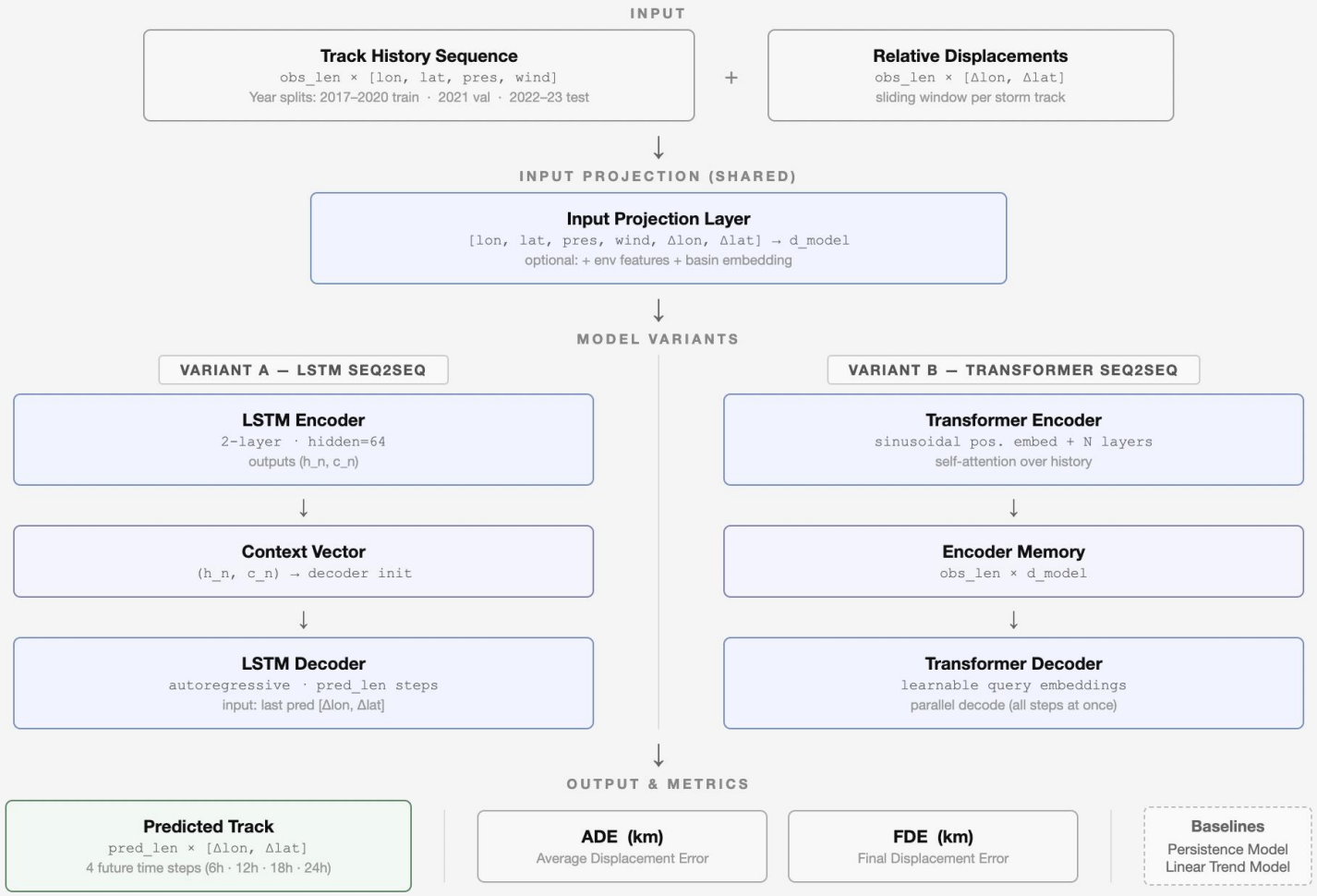
Full Dataset – SI Basin

Findings



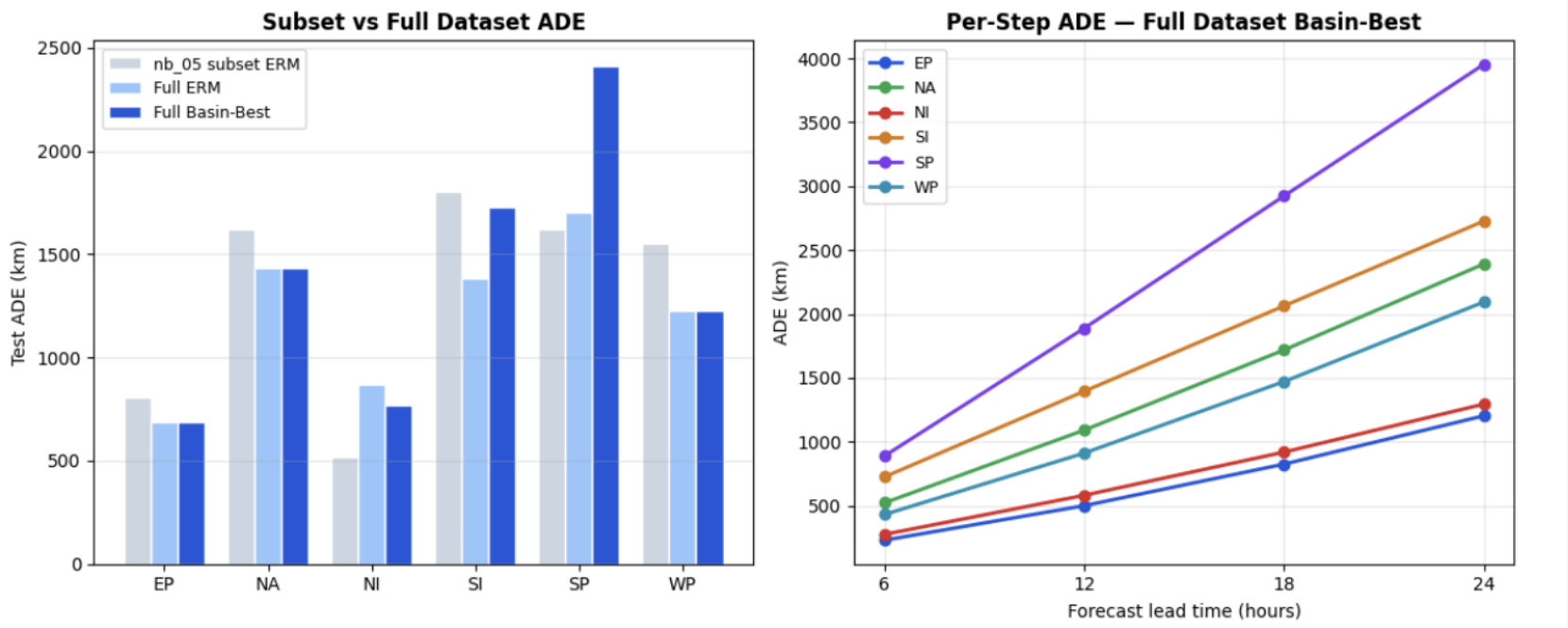
- **TrainData delivers slight improvements over TestData:** IntAcc: 0.560 → 0.575 (+2.7%), WindMAE: 5.09 → 4.35 m/s (−14.5%).
- **23,986 training samples, best checkpoint at epoch 5:** Val F1 peaked early (0.604 at epoch 5) and degraded by epoch 10 (0.579). Early stopping correctly restores the epoch 5 weights.
- **WindMAE 4.35 m/s is the best regression result across all experiments.**
- **Direction accuracy drops (0.567 → 0.272):** Assumption is, earlier TestData used year-split with a temporal shortcut; TrainData's strict folder split means the model must generalise, across true unseen storms. Lower DirAcc reflects a harder, cleaner evaluation — not a regression.

Our Architecture
Version 2.1



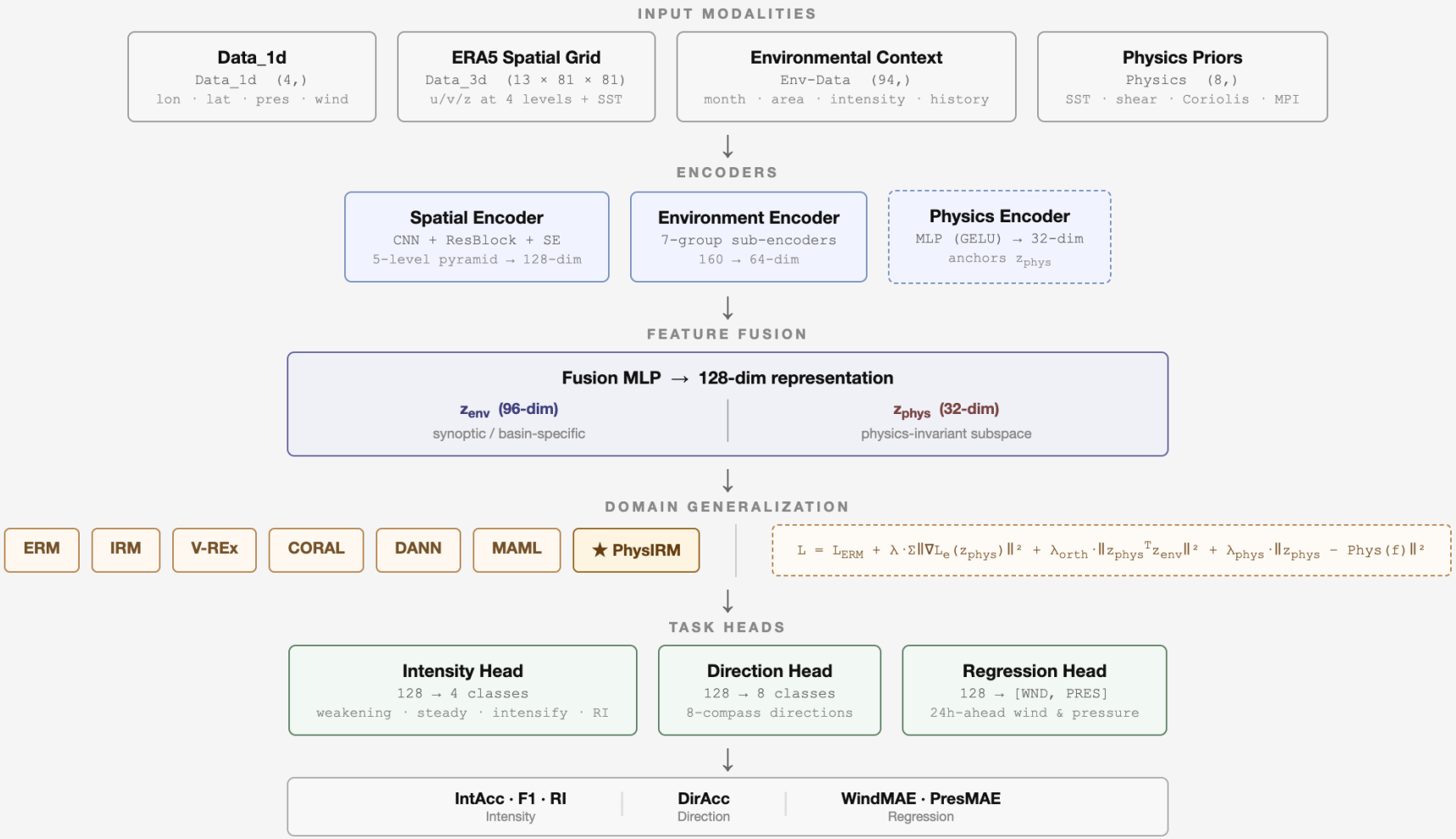
Partial Dataset vs Greedy Source vs Full Dataset

Findings

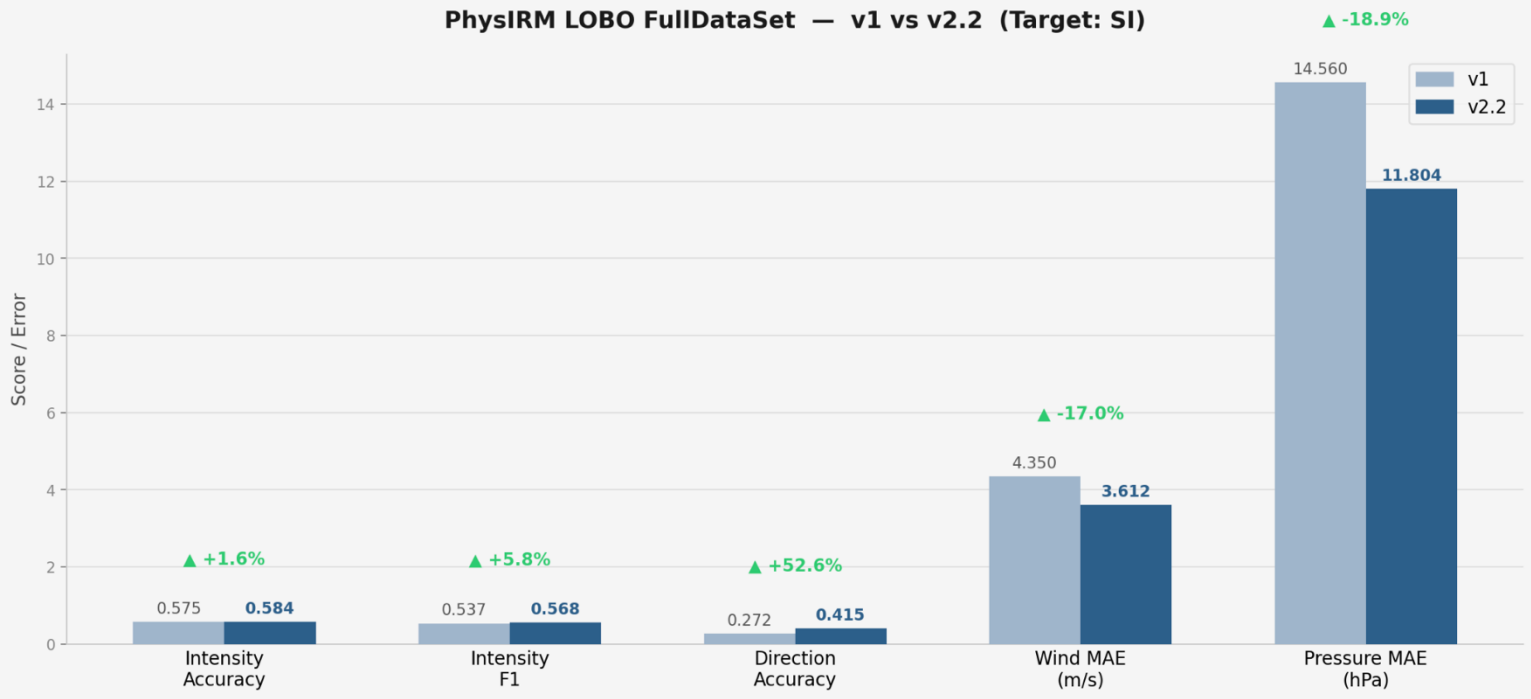


Dropping Data3D and EnvData fixes ADE: 10K km → 1K km Multimodal gave catastrophically high ADE (~10k km). Running the same LOBO with Data1D only brings mean ADE down to 1K km.

Our Architecture
Version 2.2



Comparing Results
Findings



- **Across-the-board improvement:** v2.2 wins on all 5 metrics. Every single bar moves in the right direction. No trade-off, no regression. The architectural upgrade is unambiguous.
- **Direction Accuracy is the standout gain (+52.6%):** From 0.272 → 0.415. This is the largest relative jump, showing v2.2 learns directional storm movement significantly better, the weakest point of v1 is now its biggest improvement.
- **Regression errors drop sharply:** Wind MAE falls 17% (4.35 → 3.61 m/s), Pressure MAE falls 18.9% (14.56 → 11.80 hPa). The model's physical quantity predictions are more accurate, not just its classifications
- **One model, five tasks, all better:** This is not a single-metric optimisation. v2.2 improves simultaneously on classification, multi-class direction, wind regression, and pressure regression, evidence of a stronger shared representation, not just task-specific tuning.

Future Scope

1 Scale Up

Full LOBO sweep across all basins, more epochs

2 Strong Physics

Refinement and Ablation Studies

3 Extended Lead Time

Push beyond 24h to 48h and 72h lead times.

4 Robustness Reporting

Standardize uncertainty metrics for RI events.

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Thank you

https://github.com/arshadpatel2001/MLfTCC

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